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IdeaWeb—Understanding Workplace Networks (A)

The Work Collaboration Network

Quito Duran was thrilled to be here. He was the newest hire in IdeaWeb’s London office, and he was getting a tour of the office by Risha Singh, head of the group. IdeaWeb was a product and research consultancy. The people who worked here were from diverse backgrounds, including design, software, operations, and elsewhere. The types of projects the company worked on ranged from developing new products and services to brand redesigns and public awareness programs.

As Duran settled in, he was keen to demonstrate his value, so he set out to understand the working relationships in the group. IdeaWeb divides its market into sixteen regions with one office group located per region. Each group was run by the group head. Within each group, employees were divided into teams and each team had a team leader. The team leaders typically reported to the group head.

To learn more, Duran had lunch with a few colleagues, who explained that work collaborations largely followed the organizational chart hierarchy. This surprised Duran, who thought that fluidity in working relationships would be critical at an innovative company like IdeaWeb. He did enjoy the lunches though—they were catered daily from local restaurants, and they seemed to be a key part of the company’s culture. Most employees ate lunch at the same time, and he saw many conversations taking place as he ate.

Knowing that formal relationships were salient in the office, Duran used the group’s organization chart and his understanding of working relationships to sketch out the group’s working network. A connection between two employees represents the existence a formal reporting/working relationship between the two. Duran’s network graph is produced in Figure 1. He also computed network and employee centrality measures, shown in Tables A and B, respectively.

After showing his work to Singh, she said, “I am really glad you worked on this. It turns out we have an issue with our lease agreement on this floor. We will have to relocate to two different floors in the building.

Unfortunately, the new floors are smaller—the maximum capacity on each is 15 employees. It is somewhat inconvenient because the floors are serviced by different elevator banks. So, if someone wanted to visit the other floor, they would first need to go to the ground level, and then switch elevator banks before going back up again. Can you propose a division of employees across the two floors, given your network of working relationships?”

Case Questions

As you answer these questions, keep in mind that the details of the case may not contain all the information you need to arrive at an answer. Imagine how this business would operate to come up with feasible answers to the questions.

1. The network degree density is 0.101. What does this tell you about the nature of collaborations in this group?

Degree density of 0.101 means that only about 10.1% of all possible connections between employees actually exist in the network. **This is a sparse network.**

Key implications:

- Collaboration is highly structured and concentrated: most employees only collaborate with a small, defined set of colleagues — their direct team.
- Formal hierarchy is the dominant organising principle. As colleagues confirmed to Duran at lunch, work relationships largely follow the org chart, not informal preference or expertise.
- Information flow is constrained: with only ~10% of possible links active, knowledge and ideas pass through a limited number of pathways, creating potential bottlenecks at key bridge nodes (e.g. Risha, Ed, Sujin).
- This is typical of hierarchical organisations but can inhibit cross-functional innovation — which is particularly notable at a creativity-focused consultancy like IdeaWeb.

Contrast: A density of ~0.5 would indicate a very integrated, flat team where almost everyone collaborates with everyone else. The low density here suggests IdeaWeb's formal structure is quite siloed.

2. For each respective centrality measure, which employees have rank highly? What does each kind of measure tell you about the nature of those employees' centrality? How do the measures comport with your intuition when looking at the graph?

- Top: Ed (0.222) and Sujin (0.222). Both are team leaders with the most direct connections.
- What it means: Degree centrality captures raw connectivity — who has the most direct one-to-one working relationships. High-degree employees are the primary conduits of information and workflow within their immediate neighborhood. Ed and Sujin each manage large teams, and also connect upward to Risha. Their degree reflects volume of collaboration.

Closeness Centrality — who can reach others most quickly?

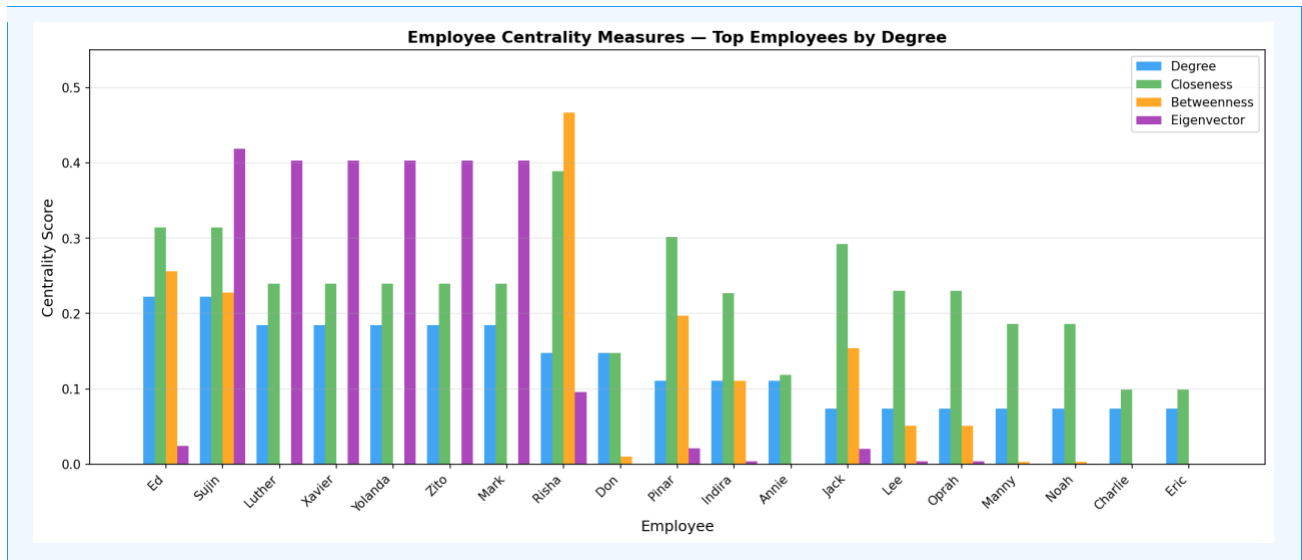
- Top: Risha (0.389), Ed (0.314), Sujin (0.314). Risha stands out despite having fewer direct connections than Ed/Sujin.
- What it means: Closeness captures how quickly someone can reach anyone else in the network, on average. Risha, as group head, sits at the top of the hierarchy. Even though she has fewer direct links, those links connect her to all team leaders, giving her short paths to every part of the organisation. This comports with intuition: a group head is well-positioned to disseminate information widely and quickly.

Betweenness Centrality — who controls the flow between others?

- Top: Risha (0.467), Ed (0.256), Sujin (0.228). Risha's score is notably dominant.
- What it means: Betweenness centrality measures how often an employee lies on the shortest path between two other employees. A high score indicates a gatekeeper or broker. Risha is the single most important conduit in the entire network — almost all cross-team collaboration must pass through her. Ed and Sujin also have meaningful scores since they bridge their teams to the rest of the organisation. Many employees (Luther, Xavier, Yolanda, etc.) have a betweenness of 0.000, meaning they lie on no shortest paths — they are "leaf" nodes in the network.
- Insight: Risha is a potential bottleneck. If she were absent or overwhelmed, information flow across the group would be severely impaired.

Eigenvector Centrality — who is connected to the most influential people?

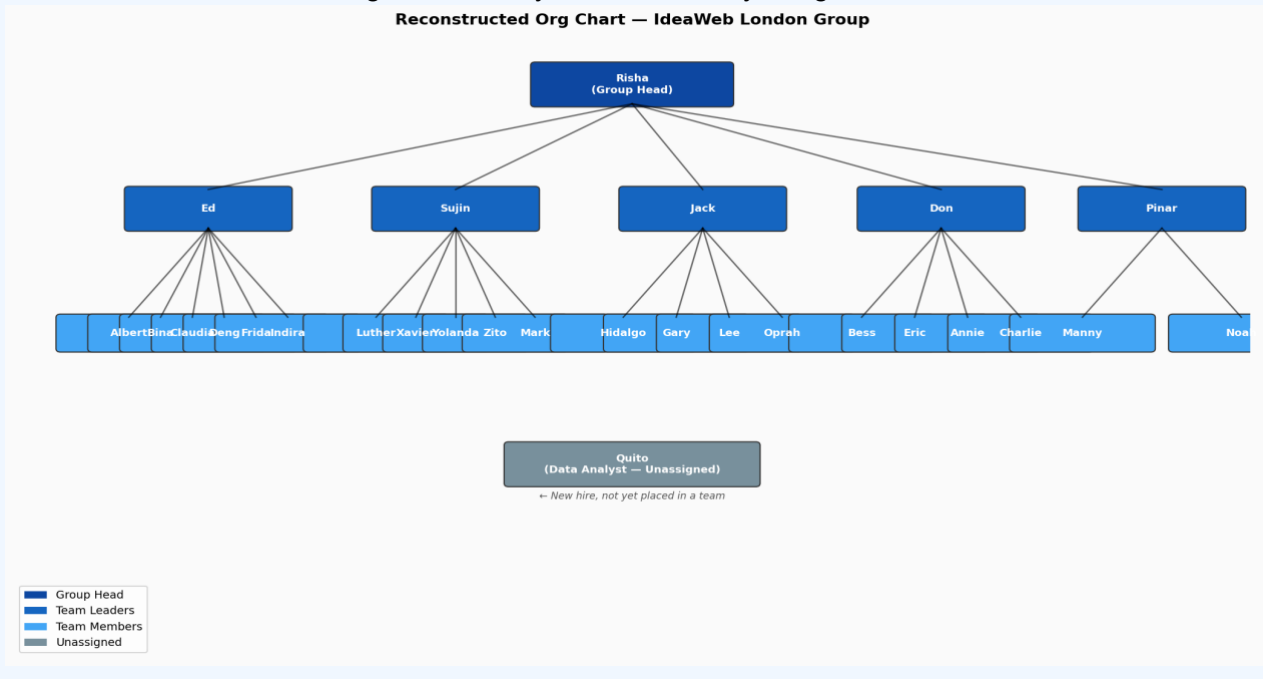
- Top: Sujin (0.419), then Luther, Xavier, Yolanda, Zito, Mark (all at 0.403). Notably, Sujin's team dominates this metric.
- What it means: Eigenvector centrality weights connections by the importance of one's connections — being linked to influential people increases your score. Sujin's high score reflects that she is both directly connected to a large team AND to Risha (a high-eigenvector node). Her team members inherit high scores because they are connected to Sujin. Intuitively, this identifies the "well-connected cluster" — the group that is most embedded in the powerful core of the network.
- Note: Ed, despite having the same degree as Sujin, has a much lower eigenvector score (0.024), because Ed's team (Albert, Bina, Claudia, etc.) have very few other connections — they are "leaf" nodes who do not amplify Ed's influence the way Sujin's team does.



- Given Duran's work collaboration network, attempt to recreate the formal organizational chart of the London group. You can sketch the chart by hand or use an online tool like diagrams.net.

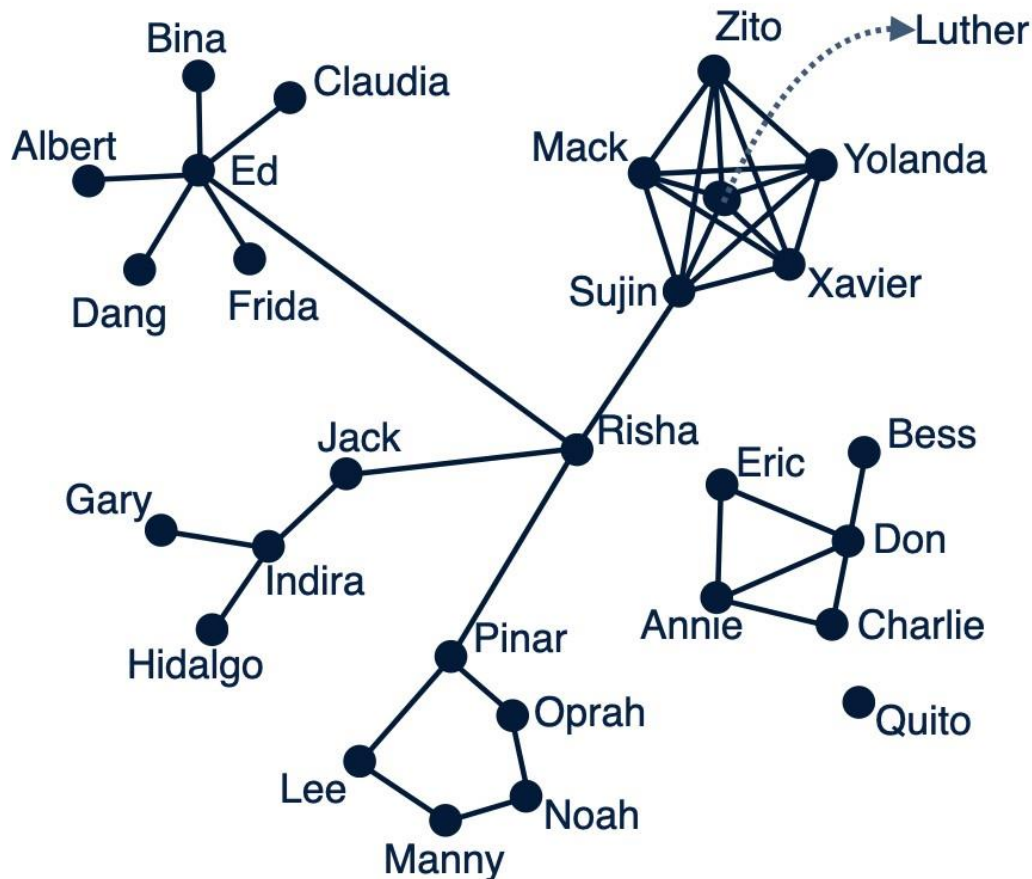
The org chart is reconstructed from the network graph and centrality data. The structure follows a clear hierarchical pattern:

- Risha Singh is the Group Head at the top of the hierarchy.
- Five team leaders report to Risha: Ed, Sujin, Jack, Don, and Pinar — each with relatively high degree and closeness centrality.
- Each team leader oversees a team of 2–6 members who appear as leaf nodes (betweenness = 0, low degree).
- Quito (the new hire) is not yet placed in any team — he shows 0 across all centrality measures, confirming he has not yet been formally integrated.



- Propose how you split the employees among the two floors (remember there is a limit of 15 employees per floor). Sketch your proposal on the network diagram below.

Figure 1 Network Graph



Proposed allocation principle:

The key objective is to minimise the number of working relationships that are severed by the floor separation (i.e., cross-floor edges should be minimised). Given the hierarchical structure, the most natural split preserves whole teams together and keeps highly-betweenness nodes (Risha, Ed) on the same floor as their key collaborators.

Floor 1 (15 employees) — Jack's, Sujin's, and Don's teams:

- Jack, Indira, Gary, Hidalgo (Jack's team — 4 employees)
- Sujin, Mark, Xavier, Yolanda, Luther, Zito (Sujin's team — 6 employees)
- Don, Bess, Eric, Annie, Charlie (Don's team — 5 employees)

Floor 2 (13 employees) — Risha's staff, Ed's, and Pinar's teams:

- Risha and Quito (Group Head + Data Analyst)
- Ed, Albert, Bina, Claudia, Dang, Frida (Ed's team — 6 employees)
- Pinar, Oprah, Noah, Manny, Lee (Pinar's team — 5 employees)

Rationale:

- This split severs only 5 cross-floor formal edges (Risha's connections to Jack, Sujin, and Don), the minimum possible given the tree-like structure.
- All teams remain intact on the same floor, preserving the most operationally critical links.

- Risha being on Floor 2 means she can still closely coordinate with Ed and Pinar. Cross-floor communication with Floor 1 team leaders (Jack, Sujin, Don) will require more effort but is manageable given Risha's formal authority.

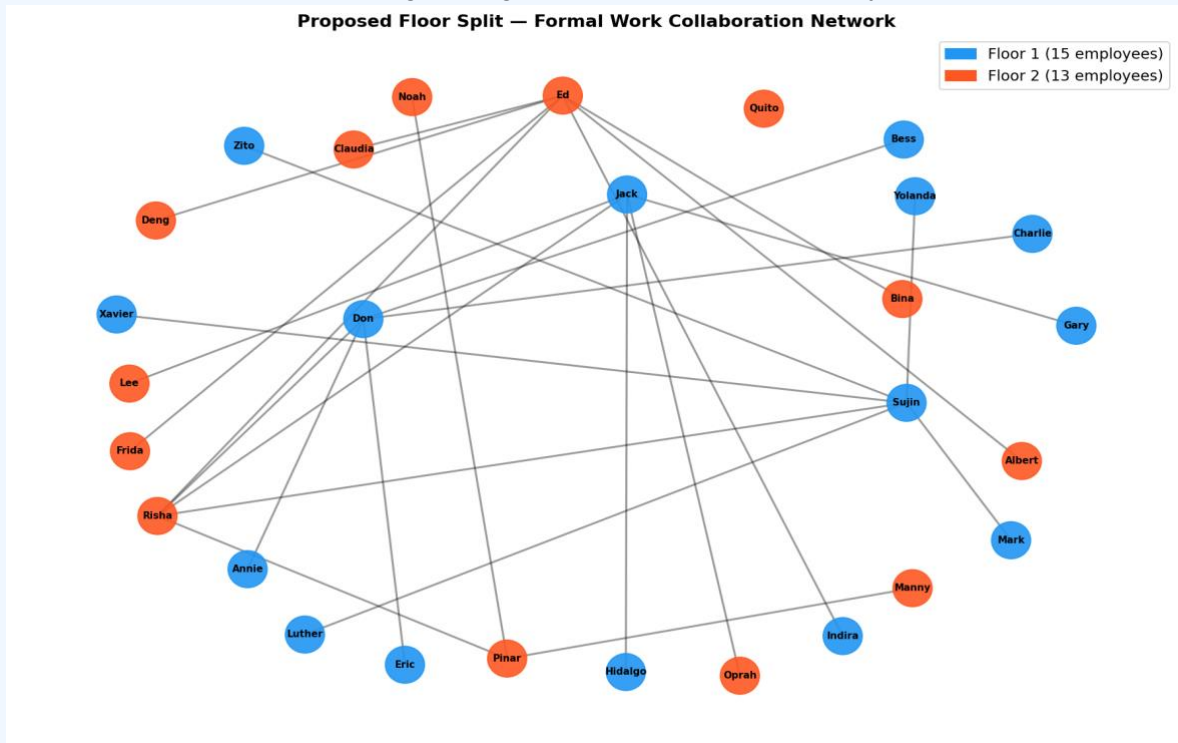


Table A Network Statistics

Category	Statistic	Value
Density	Degree Density	0.101
Distance	Diameter (largest component)	6
Distance	Average Distance (largest component)	3.312
Clustering	Overall Clustering	0.611
Clustering	Average Clustering	0.310

Table B Employee Centrality Statistics (sorted by degree)

Name	Degree	Closeness	Betweenness	Eigenvector
Ed	0.222	0.314	0.256	0.024
Sujin	0.222	0.314	0.228	0.419
Luther	0.185	0.240	0.000	0.403
Xavier	0.185	0.240	0.000	0.403
Yolanda	0.185	0.240	0.000	0.403
Zito	0.185	0.240	0.000	0.403
Mark	0.185	0.240	0.000	0.403
Risha	0.148	0.389	0.467	0.096
Don	0.148	0.148	0.010	0.000
Pinar	0.111	0.302	0.197	0.021
Indira	0.111	0.227	0.111	0.004
Annie	0.111	0.119	0.001	0.000
Jack	0.074	0.292	0.154	0.020

Lee	0.074	0.230	0.051	0.004
Oprah	0.074	0.230	0.051	0.004
Manny	0.074	0.186	0.003	0.001
Noah	0.074	0.186	0.003	0.001
Charlie	0.074	0.099	0.000	0.000
Eric	0.074	0.099	0.000	0.000
Albert	0.037	0.227	0.000	0.005
Frida	0.037	0.227	0.000	0.005
Deng	0.037	0.227	0.000	0.005
Claudia	0.037	0.227	0.000	0.005
Bina	0.037	0.227	0.000	0.005
Hidalgo	0.037	0.178	0.000	0.001
Gary	0.037	0.178	0.000	0.001
Bess	0.037	0.085	0.000	0.000
<u>Quito</u>	<u>0.000</u>	<u>0.000</u>	<u>0.000</u>	<u>0.000</u>

IdeaWeb—Understanding Workplace Networks (B)

The Hidden Network

Duran proposed dividing placing Jack’s, Sujin’s, and Don’s teams¹ on one floor and the remaining employees on the other floor. His recommendation on the employee division across the two floors was implemented. While there was some initial disappointment from splitting the group in two, everyone appreciated the reality of the situation and they immediately adjusted to their new circumstances.

Three months after the office reorganization, Singh walked into Duran’s office. “We have an urgent matter that needs to be addressed,” she said. “It seems that since we moved to two separate floors, our clients’ overall satisfaction with our service has declined noticeably. We are seeing lower levels of satisfaction with our responsiveness and turnaround times as well as the perceived quality of our ideas. Figuring out what happened and how to remedy this is your top priority.”

Duran mulled the news. The timing of the office relocation and the subsequent decline in client satisfaction certainly suggested a causal relationship. Even if he assumed the move caused the decline in the group’s performance, he was not sure *how* it caused the decline. Did the division negatively affect employee morale, regardless of the specific allocation? Or, did it relate to the specific division of employees were divided across the two floors?

Duran always believed that fluid interactions were a critical part of the creative process that fueled IdeaWeb. He hypothesized that dividing the employees across the two floors based on formal working groups must have severed several *informal* relationships that were essential to the group’s performance. He thought of one in particular—the daily lunch, where employees sat with colleagues from other areas and they had an opportunity to converse about projects and clients.

To begin his assessment of this hypothesis, Duran collected a second network dataset. This time, he asked everyone to name all the colleagues they had lunch with (at least twice) during the six-month period prior to the office relocation. Duran’s graph of the lunch network is shown in Figure 2.

Case Questions

1. Why is the timing of the employee division and the subsequent decline in client satisfaction not sufficient to conclude a causal relationship?

This is the classic problem of correlation vs. causation. Even if two events happen sequentially, that does not mean the first caused the second. Several alternative explanations could account for the simultaneous timing:

- Confounding variables: Something else may have changed at the same time as the floor move — a change in management style, loss of a key client-facing employee, shifts in the client base or market conditions — that independently caused the satisfaction decline.
- Reverse causality: Although unlikely here, if satisfaction was already declining, this could have triggered organisational changes (like the move), not the other way around.
- Regression to the mean: If satisfaction was unusually high before the move, it might have naturally reverted toward average levels regardless of any intervention.
- Latent trends: Satisfaction may have been on a downward trajectory before the move, with the change simply coinciding with a pre-existing trend.

To establish causality, Duran would ideally need: a control group (another office that did not move), evidence ruling out confounders, and/or a deeper analysis of the mechanism (the "how") through which the move affected performance.

2. If we assume the relationship is causal, why is it important to know how the division caused the decline?

Understanding the causal mechanism is critical for designing an effective remedy. Without knowing the "how," any intervention is essentially guesswork. Consider two very different causal pathways:

- Pathway 1 — General morale: The act of physically separating colleagues, regardless of which employees ended up on which floor, damaged morale and cohesion. If this is the cause, the solution would involve cultural interventions (all-hands meetings, joint events, cross-floor rituals) rather than changing the floor assignments.
- Pathway 2 — Specific network disruption: The particular way employees were assigned to floors severed key informal relationships that were driving creative performance. If this is the cause, the solution is to reassign employees across floors to restore those critical connections.

These two remedies are completely different. Misidentifying the mechanism would lead to wasted resources on an ineffective fix. Furthermore, knowing the mechanism reveals which relationships and interactions are truly "load-bearing" for performance — a valuable insight for future organisational design decisions.

3. What are your observations from the lunch network shown in Figure 2?

- Much denser than the formal network: The lunch network contains far more connections, suggesting employees engage informally with a much wider range of colleagues than their direct work teams. This is consistent with the open lunch culture described in the case.
- Cross-team connections: Many nodes are connected across team boundaries — people from different formal teams regularly lunched together. These cross-cutting

ties are exactly the kind of "weak links" (Granovetter, 1973) that bridge structural holes and facilitate the sharing of novel information.

- Different centrality landscape: Employees who had low centrality in the formal network may have high centrality in the lunch network. This means the informal network reveals a different, complementary power structure within the group.
- Quito is likely now connected: As a new hire who ate lunch with colleagues, Quito may appear as a node in the lunch network even though he is isolated in the formal work network. This illustrates how informal integration can precede formal integration.
- Floor split implications: Comparing the lunch network to the floor assignments reveals which informal connections were severed by the relocation. Given that lunch was shared across the whole office (all employees ate at the same time), these cross-floor informal links are likely the ones most disrupted.

4. Based on the lunch network, how might you change the split of employees across the two floors?

The lunch network reveals which informal connections are most valuable. The revised split should apply a community detection or modularity-maximisation approach to the lunch network — partitioning employees into two groups of ≤ 15 such that cross-group lunch links are minimised.

- Identify densely connected clusters: Employees who lunch together frequently likely share project ideas, client insights, and creative energy. These clusters should be kept on the same floor.
- Preserve bridge nodes: If certain employees act as brokers between two large informal clusters (high betweenness in the lunch network), consider which cluster they are more densely embedded in and place them accordingly. Alternatively, if a broker connects two equally important groups, their floor placement becomes the critical decision.
- Re-examine Risha's position: Risha's placement on Floor 2 in the formal split may not reflect her role in the informal network. If she has strong informal ties to Floor 1 employees, moving her to Floor 1 (or choosing a different split entirely) could help.

Operationally, Duran should: (1) compute community structure of the lunch network using algorithms such as Louvain or Girvan-Newman, (2) identify the two-partition that minimises cross-floor informal ties, (3) check feasibility (≤ 15 per floor), and (4) propose this revised assignment to Risha.

5. Figure 2 is an example of an informal organizational network that was based on lunch interactions. What other kinds of informal networks exist among employees in an organization and how might you collect data to create these networks?

Types of informal networks:

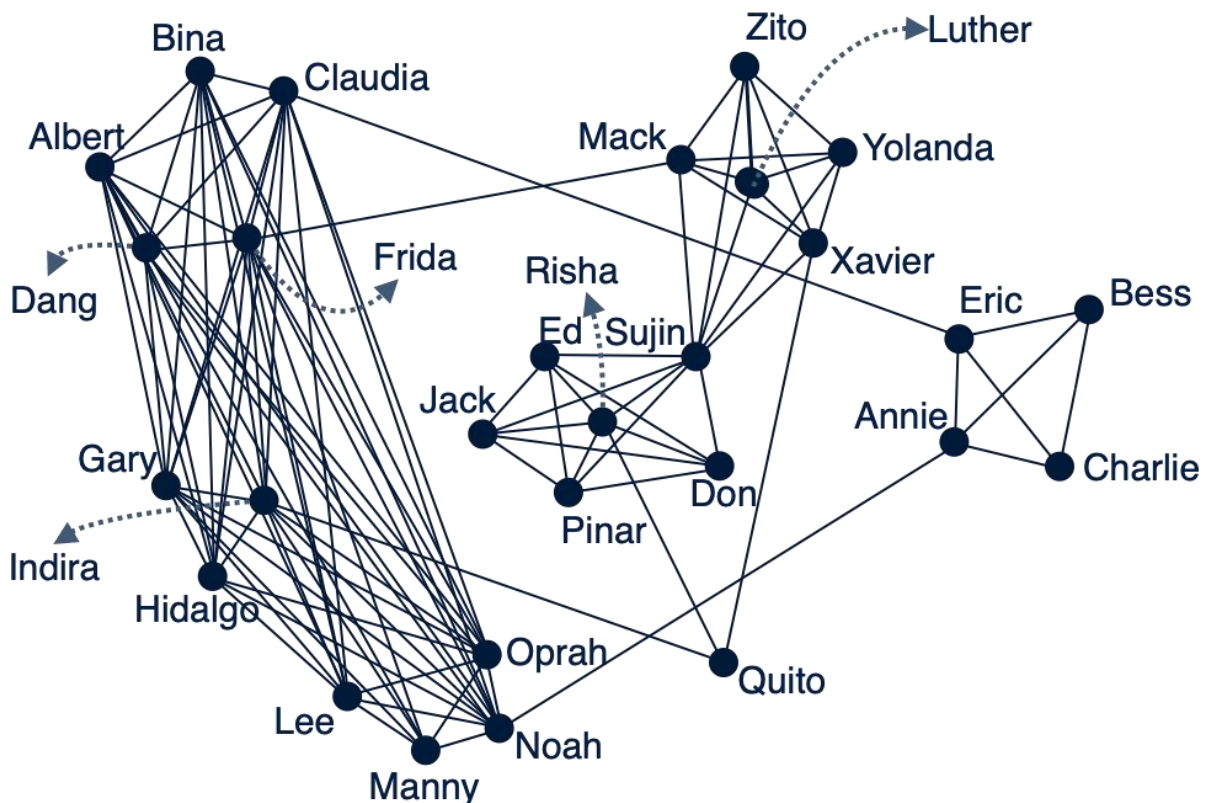
- Advice networks: Who do employees go to when they need guidance on a difficult problem? Captures expertise-seeking behaviour. Collected via survey ("Name up to 5 people you turn to for professional advice").
- Trust networks: Who do employees trust with sensitive information or personal concerns? Reveals deep interpersonal bonds. Collected via confidential survey ("Who in this organisation do you trust most?").
- Friendship networks: Social relationships outside work tasks. Collected via survey or by observing after-work activities.

- Communication networks: Who emails, messages, or calls whom? Collected (with consent) from email metadata, Slack/Teams logs, or calendar data. Objectively measurable with no self-report bias.
- Energy networks: Who energises vs. de-energises employees? Collected via survey ("After interacting with this person, do you feel more or less motivated?"). Research by Cross et al. (2003) shows this predicts performance outcomes.
- Innovation/idea networks: Who do people share new ideas with first? Collected via survey or by tracking which informal conversations precede project proposals.

Data collection methods:

- Surveys & sociometric questionnaires: Most common; can target specific relationship types. Risk: respondents may under-report or over-report connections.
- Digital traces: Email/Slack metadata, calendar invites, badge/RFID proximity data. Objective but raises privacy concerns and requires employee consent.
- Observations and shadowing: A researcher observes who talks to whom, in which settings. Accurate but resource-intensive and may cause observer bias.
- Passive sensing / wearables: Sociometric badges track face-to-face interactions, voice patterns, and physical proximity. High-resolution but potentially invasive.

Figure 2 Visualization of Employee Lunch Network



IdeaWeb—Understanding Workplace Networks (C)

Extending the Analysis

Duran's recommendations on how to recover the high levels of customer satisfaction the company was known for proved to be successful. Clients quickly took notice and satisfactions improved nearly to pre-move levels. Singh was thrilled, "Thank you for sorting out what could have been a disaster for our group. It is good to know that the connections among our staff are not always represented by the organization chart. I would like you to expand your analysis to the entire company."

Duran's first step in this new exercise was to collect the collaboration (formal) and lunch (informal) networks for the other fifteen regional groups. There were 438 employees, including the London office staff. He decided to look at whether there were any relationships between the centrality measures and performance-related outcomes. The data for both outcomes were collected in the three-month period following the network data and included (a) whether the employee was promoted and (b) whether the employee's next 360^a performance review¹ was negative. Duran also coded a few additional variables, including time at the company (in years), and dummy variables for gender, work travel frequency, and recent promotion.

With his employee-level dataset and two outcomes of interest, Duran ran two regressions. The regression results are presented in Table C. Each column represents a single regression. In Column 1, the dependent variable is *promoted*, which equals 1 if the employee was promoted within three months after the collection of the network data, and 0 otherwise. In Column 2, the dependent variable is *negative review*, which equals 1 if or the employee received an overall negative score in the performance review following the collection of the network data, and 0 otherwise. Each centrality variable is normalized (i.e. the value from each observation is subtracted by the variable mean and divided by the variable standard deviation), so coefficients on those variables would be interpreted as the effect size associated with a one standard deviation increase in the respective centralization measure

Duran peered at the results. There were some interesting findings he would write up in a memo to company management and he already conceived of additional steps he might pursue to learn more.

Case Questions

1. What are your takeaways from the two regressions? Point out specific coefficients (and interpret them) when making your points.
2. The coefficient of informal degree centrality is negative and statistically significant in the *negative review* regression. Based on this result, how do you view holding more social events to promote informal connections between employees, which would result in a reduction of the likelihood of employees receiving negative reviews?
3. What are some limitations of this analysis?
4. What additional regressions would you consider running? What next steps might you pursue after looking at this analysis?

Table C Regression Results for Employee Outcomes

Dependent variable:	(1) promoted	(2) negative review
Formal network centrality		
Degree	-0.015 (0.025)	0.021* (0.009)
Closeness	0.004 (0.146)	-0.035** (0.013)

Betweenness	0.113**	-0.029
	(0.042)	(0.019)
Eigenvector	0.084***	-0.091**
	(0.025)	(0.031)
Informal network centrality		
Degree	-0.039	-0.011***
	(0.030)	(0.001)
Closeness	0.085+	-0.036*
	(0.051)	(0.018)
Betweenness	0.103	-0.001
	(0.742)	(0.001)
Eigenvector	0.102*	-0.084
	(0.046)	(0.055)
Years at company	-0.008***	-0.01*
	(0.000)	(0.004)
Female	-0.12***	0.168***
	(0.002)	(0.009)
High frequency traveler	0.048	0.033**
	(0.102)	(0.012)
Promoted in last 12 months	-0.041*	-0.044***
	(0.020)	(0.012)
<u>Regional Group Dummies</u>	<u>Yes</u>	<u>Yes</u>
Employees	438	438
<u>R-squared</u>	<u>0.473</u>	<u>0.348</u>

Note: + $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Each column of this table presents results from a single OLS regression. The dependent variable is *promotion* and *negative review* in Columns 1 and 2, respectively. The included independent variables are four measures of employee centrality for both the formal and informal networks, additional control for employee characteristics and behaviors, and dummy variables for each regional group (results not shown). For each variable, the coefficient and robust standard error (in parenthesis) is shown. The constant is omitted.

Across 438 employees, Table C reports two OLS linear probability models with robust standard errors ($R^2 = 0.473$ for promotion; $R^2 = 0.348$ for negative review). Network centralities are standardised ($SD = 1$), so each coefficient is the ceteris paribus percentage-point change in the outcome associated with a one standard deviation increase in that centrality measure, holding the other network measures, employee controls, and regional-group dummies constant. Promotion equals 1 if the employee is promoted within three months after network collection; negative review equals 1 if the next 360° review after network collection is negative.

Promotion is primarily associated with *positional advantage in the formal network*, not raw connectedness. Formal betweenness is the strongest network predictor (+0.113, $p < 0.01$), implying an 11.3 pp higher promotion probability per SD – consistent with brokerage roles that connect otherwise separate parts of the collaboration structure. Formal eigenvector centrality is also strongly positive (+0.084, $p < 0.001$; +8.4 pp), indicating that proximity to influential, well-connected formal colleagues is associated with advancement. In contrast, formal degree is not significant (−0.015), and formal closeness is effectively zero (0.004), reinforcing that “more ties” or “shorter paths” alone do not predict promotion once brokerage and prestige are accounted for. Informal effects are weaker and more selective: informal eigenvector is positive (+0.102, $p < 0.05$; +10.2 pp) and informal closeness is only marginal (+0.085, $p < 0.10$), while informal degree and informal betweenness are not reliable predictors.

For negative reviews, the pattern shifts toward *social and status-like buffering* with one exposure effect. Formal degree becomes positive and significant (+0.021, $p < 0.05$; +2.1 pp), consistent with greater evaluative exposure in larger formal working networks. However, the dominant formal associations are protective: formal eigenvector is large and negative (−0.091, $p < 0.01$; −9.1 pp) and formal closeness is negative (−0.035, $p < 0.01$; −3.5 pp), while formal betweenness is not significant. Informal embeddedness also protects: informal degree is negative and highly significant (−0.011, $p < 0.001$; −1.1 pp) and informal closeness is negative (−0.036, $p < 0.05$), whereas informal eigenvector and betweenness are not significant.

Key covariates materially condition the story: tenure reduces promotion (−0.008, $p < 0.001$), high-frequency travel increases negative-review risk (+0.033, $p < 0.01$), and recent promotion lowers both subsequent promotion (−0.041, $p < 0.05$) and negative reviews (−0.044, $p < 0.001$). Most strikingly, female employees face a 12.0 pp promotion deficit and a 16.8 pp higher negative-review probability (both $p < 0.001$), magnitudes that exceed most network coefficients and warrant separate managerial investigation beyond network-based interventions.